# Exam, Macroeconomics III

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**Instructions:** Answer all questions clearly and concisely. Show all your work where applicable. Explain intuition when asked.

## Exercise 1 – Multiple Choice questions

Please answer the following multiple choice questions. For each question, write the question number and only the letter of the correct answer: "A", "B", "C", or "D". There is always exactly one correct answer. If you give more than one letter for a question, you receive zero points for that question. There are no negative points for wrong answers.

1. In empirical macroeconomics, how can the slope of the Phillips Curve (relationship between inflation and real activity) be identified?

   A. By regressing inflation on output (or unemployment) using all available time-series data.

   B. By exploiting exogenous variation in aggregate demand that moves output while holding supply conditions and the Phillips Curve itself fixed, and controlling for inflation expectations.

   C. By focusing on episodes with large cost-push (supply) shocks, since these shocks shift the Phillips Curve and directly reveal its slope.

   D. By comparing long-run averages of inflation and unemployment across countries, assuming all structural differences are negligible.

   Correct answer: B) demand shocks

2. Consider a time series where we observe that the frequency of price changes rises when inflation is higher. With which type of micro-level price rigidity is this empirical pattern in line?

   A. Taylor pricing

B. Fischer contracts

C. Menu costs

D. Calvo pricing

Correct answer: C) Menu costs

3. What are local projections?

   A. A method that filters a time series using only nearby observations to estimate its long-run trend.

   B. A structural method that maps regional outcomes to aggregate outcomes using a general equilibrium model.

   C. A method that estimates impulse responses (dynamics) of an outcome variable by running separate regressions for each forecast horizon on current and lagged shocks.

   D. A forecasting approach that combines many local AR(1) models into a single global prediction for long-run outcomes.

   Correct answer: C)

4. In the standard DMP search and matching model, what is the effect of an increase in the job-finding rate for workers on the steady-state unemployment rate?

   A. It raises steady-state unemployment because firms hire fewer workers.

   B. It lowers steady-state unemployment because more unemployed workers match with vacancies faster.

   C. It has no effect on steady-state unemployment, only on job turnover.

   D. It lowers unemployment temporarily, but the long-run unemployment rate is unchanged due to wage rigidity.

   Correct answer: B)

5. Which of the following statements is true about the overlapping generations model discussed in class?

   A. The decentralized equilibrium in the model is never Pareto efficient, the planner can always bring about a Pareto improvement.

B. The first welfare theorem holds in the OLG model.

C. The OLG model implies that capital accumulation is at to the social optimum, but consumption is not.

D. Under certain parameter restrictions a Pareto improvement is possible relative to the decentralized equilibrium.

Correct answer: D)

6. In the small open economy lecture example, households consume two goods:

   - Nutella: a tradable good that can be traded internationally,
   - Croissants: a non-tradable good that must be consumed locally.

   Endowments of both goods are initially fixed. Consider a country that receives a permanent increase in its endowment of Nutella (the tradable good), while the croissant endowment is unchanged in the short run. Assume Nutella and croissants are complements in utility.

   Which of the following statements best describes the short-run effect on prices and the real exchange rate in this country?

   A. The price of croissants falls, making life cheaper; the real exchange rate depreciates because the country can now export more Nutella.

   B. The price of croissants rises because higher income from Nutella increases demand for non-tradables while their supply is fixed; the real exchange rate appreciates as life becomes more expensive in the Nutella-rich country.

   C. Both Nutella and croissant prices rise proportionally, so the real exchange rate is unchanged.

   D. Only the price of Nutella changes, since croissants are non-tradable; therefore the real exchange rate must depreciate.

   Correct answer: B)

# Exercise 2 – Ramsey model and longer lives

Consider an economy described by the Ramsey model. The total utility of the representative agent is given by

$$U = \sum_{t=0}^{\infty} \beta^t \gamma^t \frac{c_t^{1-\sigma}}{1-\sigma} \tag{1}$$

where $0 < \beta < 1$ is the discount factor and $\sigma > 1$ measures risk aversion. The parameter $0 < \gamma < 1$ governs the representative agent's exogenous "survival probability" – each period, there is a small change $(1-\gamma)$ that the agent does not survive to the next period. It is useful when trying to study the impact of longer life-expectancy on economic outcomes.

The agent's budget constraint is given by

$$a_{t+1} = a_t(1 + r_t - \delta) + w_t - c_t + \Pi_t \tag{2}$$

where $a$ represents savings, $w$ is the wage and $c$ is the agent's chosen consumption. The return on savings is made up from the interest rate $r$ and lessened by the depreciation rate $0 < \delta < 1$. The firms profits $\Pi$, if there are any, are rebated to the consumer as a lump sum payment.

a) Write the Lagrangian for the households' optimization problem, find the first order conditions that characterize the behavior of households, and derive the Euler equation (the Keynes Ramsey rule). How does an increase in the interest rate affect consumption growth? How does it affect consumption in period $t$?

b) All else equal, what does a rise in the survival probability $\gamma$ imply for consumption growth? Provide an economic explanation.

In the economy, profit maximizing firms use capital and labor to produce output. The market is perfectly competitive. The production function is given by

$$Y_t = K_t^{\alpha} L_t^{1-\alpha} \tag{3}$$

where $Y_t$ is total output, $K_t$ is capital used by the firm and $L_t$ is labor, which is constant; the parameter $0 < \alpha < 1$ is the output elasticity of capital. Firms rent capital and labor at competitive prices $r_t$ and $w_t$, respectively.

c) Derive the first order conditions of the firm's maximization problem. Show that,

when expressing variables in per-capita terms, i.e., $k = K/L$, it is true that

$$r_t = \alpha k_t^{\alpha-1} \quad \text{and} \quad w_t = (1-\alpha)k_t^{\alpha} \tag{4}$$

d) One of the so-called Kaldor facts is that the labor income share, i.e., the share of all income (output) paid as wages to workers, is constant in many countries. Show that this is the case here.

e) Show, by imposing market clearing, that the resource constraint of the economy can be written as

$$k_{t+1} = (1-\delta)k_t + k^{\alpha} - c_t \tag{5}$$

Explain what this constraint implies about the tradeoff the household faces in allocating output towards consumption and investment.

f) Draw a phase diagram with capital on the x-axis and consumption on the y-axis. Using the Euler equation and the resource constraint, mathematically find equations that describe the points at which either (i) capital is constant or (ii) consumption is constant. Clearly mark these in the diagram. Explain, using economic intuition, how both capital and consumption evolve in an economy with very high levels of $c$ and $k$.

g) Mathematically compute the steady state level of capital in the model. Also, indicate the steady state in your phase diagram. Explain, using your diagram and economic reasoning, how an economy that starts out with a very low level of capital transitions to the steady state. What additional restrictions need to be put on the model's solution to guarantee convergence to the steady state? What precisely do they rule out? Explain.

h) Figure 1 shows a strong negative correlation between the interest rate and life expectancy in the US. Show mathematically what the model above predicts about the relationship between the steady state interest rate and longevity. Explain using economic intuition.

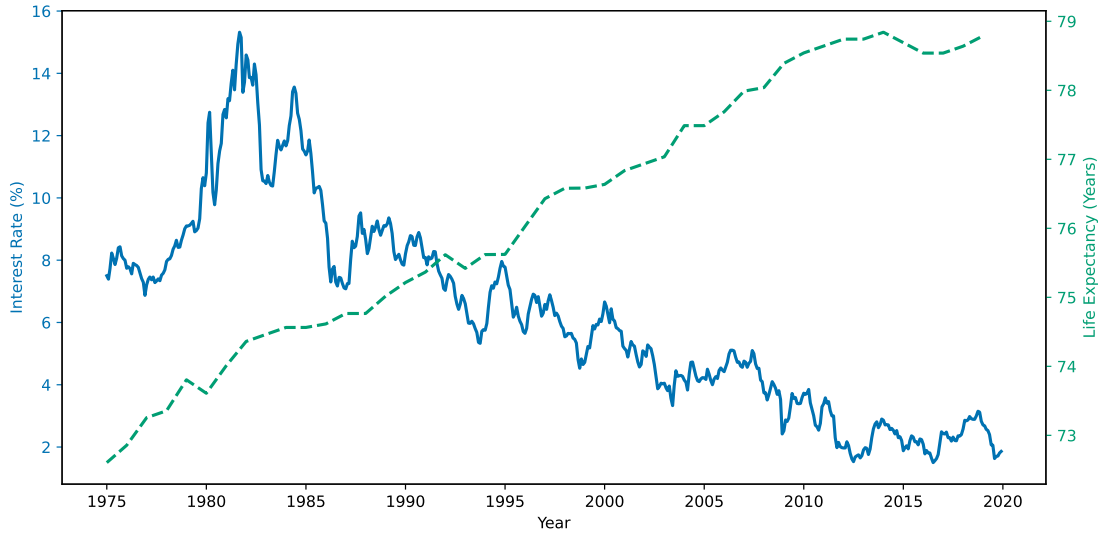



Figure 1: US Interest Rate and Life Expectancy (1975–2019)

# Exercise 3 – New Keynesian Phillips Curve

Consider a continuum of monopolistically competitive firms indexed by $i \in [0, 1]$ that face Calvo price rigidities. Time is discrete. In each period $t$:

- With probability $\alpha \in (0, 1]$ a firm is allowed to reset its price.
- With probability $1 - \alpha$ a firm must keep its price from the previous period.

All firms that can reset in period $t$ choose the same (log) price $x_t$. Let $p_t$ denote the (log) aggregate price level, $\pi_t \equiv p_t - p_{t-1}$ the (log) inflation rate, and $p_t^*$ the (log) *flexible-price* optimal price that would prevail in the absence of nominal rigidities.

The aggregate price level in this Calvo environment is

$$p_t = \alpha x_t + (1 - \alpha) p_{t-1}.$$

When a firm gets to reset its price in period $t$, this price will on average remain in place for several periods because of the Calvo friction. In log-linear approximation, the reset price $x_t$ is chosen according to

$$x_t = \frac{\sum_{j=0}^{\infty} \beta^j (1 - \alpha)^j \mathbb{E}_t \left[ p_{t+j}^* \right]}{\sum_{j=0}^{\infty} \beta^j (1 - \alpha)^j}, \tag{6}$$

where $\beta \in (0,1)$ is the discount factor.

From the underlying flexible-price model, take as given that

$$p_t^* - p_t = (\varphi - 1)y_t, \tag{7}$$

where $y_t$ is (log) output (or the output gap) and $\varphi > 1$ is the inverse Frisch elasticity of labor supply.

a) Using the building blocks above (Calvo price index, optimal reset price, and the flexible-price condition), derive the log-linear New Keynesian Phillips Curve

$$\pi_t = \kappa y_t + \beta \mathbb{E}_t[\pi_{t+1}],$$

and express the slope $\kappa$ as a function $\kappa(\alpha, \beta, \varphi)$.
*Hint:* Use the geometric series and rewrite the optimal reset price in recursive form. Then use the inflation dynamics equation to bring in inflation. Finally, use the equation that connects the (log) *flexible-price* optimal price with output.

b) Briefly discuss how the slope $\kappa$ depends on $\alpha$, $\beta$, and $\varphi$ (no detailed algebra required). Provide economic intuition.

# Suggested solutions

## Exercise 2

(a) The Lagrangian is

$$\mathcal{L} = \sum_{t=0}^{\infty} \left[ \beta^t \gamma^t \frac{c_t^{1-\sigma}}{1-\sigma} + \lambda_t \big( a_t(1 + r_t - \delta) + w_t + \Pi_t - c_t - a_{t+1} \big) \right].$$

FOCs:

$$\frac{\partial \mathcal{L}}{\partial c_t}: \qquad \beta^t \gamma^t c_t^{-\sigma} - \lambda_t = 0,$$

$$\frac{\partial \mathcal{L}}{\partial a_{t+1}}: \qquad -\lambda_t + \lambda_{t+1}(1 + r_t - \delta) = 0.$$

Combining gives the Euler equation:

$$\frac{c_{t+1}}{c_t} = \left[\beta\gamma(1 + r_t - \delta)\right]^{1/\sigma}.$$

A higher interest rate raises the return to saving, increasing consumption growth. The effect on current consumption is ambiguous because substitution (consume less today) and income (consume more today) effects work in opposite directions.

(b) From the Euler equation,

$$\frac{c_{t+1}}{c_t} = \left[\beta\gamma(1 + r_t - \delta)\right]^{1/\sigma},$$

so an increase in $\gamma$ raises consumption growth.

Higher survival probability makes households effectively more patient, raising the value of future consumption and increasing saving.

(c) Firms maximize

$$\Pi = K^{\alpha}L^{1-\alpha} - rK - wL.$$

FOCs:

$$\begin{aligned} \frac{\partial \Pi}{\partial K}: &\qquad r_t = \alpha K^{\alpha-1}L^{1-\alpha}, \\ \frac{\partial \Pi}{\partial L}: &\qquad w_t = (1-\alpha)K^{\alpha}L^{-\alpha}. \end{aligned}$$

In per-capita terms $k = K/L$:

$$r_t = \alpha k^{\alpha-1}, \qquad w_t = (1-\alpha)k^{\alpha}.$$

(d) Labor income share:

$$\frac{w_t L_t}{Y_t} = \frac{(1-\alpha)K^{\alpha}L^{1-\alpha}}{K^{\alpha}L^{1-\alpha}} = 1 - \alpha,$$

which is constant.

(e) starting from the budget constraint, we have

$$a_{t+1} = a_t(1 + r_t - \delta) + w_t - c_t + \Pi_t$$

imposing market clearing implies

$$k_{t+1} = k_t(1 + r_t - \delta) + w_t - c_t + \Pi_t$$

Plugging in firm profits and simplifying yields

$$\begin{aligned} k_{t+1} &= k_t(1 + r_t - \delta) + w_t - c_t + k_t^\alpha - k_t r_t - w_t \\ k_{t+1} &= k_t(1 - \delta) - c_t + k_t^\alpha \end{aligned}$$

Output per worker $k^\alpha$ is allocated to consumption and investment. Investment must cover depreciation $\delta k_t$ and build new capital. Thus consuming more today leaves less to accumulate capital for tomorrow.

(f) From the resource constraint, $k$ is constant when

$$k_{t+1} = k_t \quad \Rightarrow \quad c = k^\alpha - \delta k.$$

From the Euler equation, $c$ is constant when

$$\beta\gamma(1 + r_t - \delta) = 1 \quad \Rightarrow \quad \alpha k^{\alpha-1} = \frac{1}{\beta\gamma} - 1 + \delta.$$

This is a vertical line in the $(k, c)$ diagram.

*Intuition:* For very high $c$, saving is too low and capital falls (move left). For very high $k$, the marginal product of capital, i.e., the interest rate, is low, reducing consumption growth (move down).

(g) Steady state ($k_{t+1} = k_t$ and $c_{t+1} = c_t$) satisfies

$$\alpha(k^*)^{\alpha-1} = \frac{1}{\beta\gamma} - 1 + \delta.$$

Thus

$$k^* = \left( \frac{\alpha}{\frac{1}{\beta\gamma} - 1 + \delta} \right)^{1/(1-\alpha)}.$$

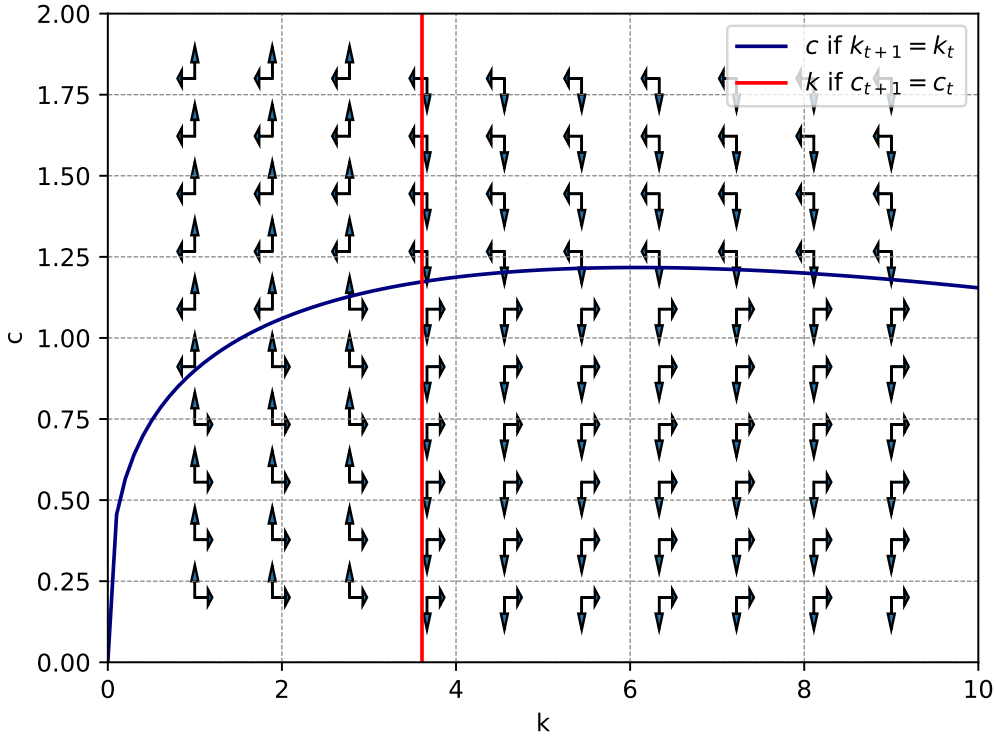



Figure 2: Phase diagram of the Ramsey model

Consumption in steady state:

$$c^* = (k^*)^\alpha - \delta k^*.$$

The steady state is where desired consumption growth is zero and investment just offsets depreciation. Any model starts with a given initial level of capital, $k_0$. From this level of capital, only a single point $c_0$ will allow the model to converge to the steady state – all others lead to explosive dynamics. To rule out these explosions, in addition to the first order conditions and constraints mentioned so far, we require a transversality and a no-Ponzi game condition. The former rules out excessive savings (bottom right) in Figure 2 while the latter rules out excessive consumption (top left).

(h) In steady state,

$$r^* = \alpha(k^*)^{\alpha-1} = \frac{1}{\beta\gamma} - 1 + \delta.$$

Differentiating,

$$\frac{\partial r^*}{\partial \gamma} = -\frac{1}{\beta\gamma^2} < 0.$$

A higher survival probability makes households more patient, they save more, capital rises, and the equilibrium interest rate falls.

## Exercise 3

## (a) Deriving the New Keynesian Phillips Curve

**Step 1: Inflation and the reset price.** From the price level equation

$$p_t = \alpha x_t + (1-\alpha)p_{t-1}$$

subtract $p_{t-1}$:

$$p_t - p_{t-1} = \alpha x_t + (1-\alpha)p_{t-1} - p_{t-1} = \alpha(x_t - p_{t-1}).$$

By definition, $\pi_t \equiv p_t - p_{t-1}$, hence

$$\pi_t = \alpha(x_t - p_{t-1}). \tag{8}$$

**Step 2: Rewrite the reset price in recursive form.** The denominator of (6) is a geometric series:

$$\sum_{j=0}^{\infty} \beta^j (1-\alpha)^j = \sum_{j=0}^{\infty} [\beta(1-\alpha)]^j = \frac{1}{1-\beta(1-\alpha)},$$

so

$$x_t = (1-\beta(1-\alpha)) \sum_{j=0}^{\infty} \beta^j (1-\alpha)^j \mathbb{E}_t[p^*_{t+j}].$$

Splitting the sum into the $j=0$ term and $j \geq 1$,

$$x_t = (1-\beta(1-\alpha))p^*_t + (1-\beta(1-\alpha)) \sum_{j=1}^{\infty} \beta^j (1-\alpha)^j \mathbb{E}_t[p^*_{t+j}].$$

Re-indexing the second sum and using the same expression one period ahead, one obtains the standard recursive representation

$$x_t = (1-\beta(1-\alpha))p^*_t + \beta(1-\alpha)\, \mathbb{E}_t[x_{t+1}]. \tag{9}$$

**Step 3: From the reset price to inflation dynamics.** Subtract $p_t$ from both sides of (9):

$$x_t - p_t = (1-\beta(1-\alpha))(p^*_t - p_t) + \beta(1-\alpha)\, \mathbb{E}_t[x_{t+1} - p_t].$$

Using the Calvo price level equation one period ahead,

$$p_{t+1} = \alpha x_{t+1} + (1-\alpha)p_t \quad \Rightarrow \quad p_{t+1} - p_t = \alpha(x_{t+1} - p_t),$$

so

$$\pi_{t+1} \equiv p_{t+1} - p_t = \alpha(x_{t+1} - p_t) \quad \Rightarrow \quad x_{t+1} - p_t = \frac{\pi_{t+1}}{\alpha}.$$

Taking expectations,

$$\mathbb{E}_t[x_{t+1} - p_t] = \frac{1}{\alpha}\, \mathbb{E}_t[\pi_{t+1}].$$

Hence

$$x_t - p_t = (1-\beta(1-\alpha))(p^*_t - p_t) + \beta(1-\alpha)\frac{1}{\alpha}\, \mathbb{E}_t[\pi_{t+1}]. \tag{10}$$

Using (8),

$$x_t - p_{t-1} = \frac{\pi_t}{\alpha},$$

so

$$x_t - p_t = (x_t - p_{t-1}) - (p_t - p_{t-1}) = \frac{\pi_t}{\alpha} - \pi_t.$$

Insert this into (10):

$$\frac{\pi_t}{\alpha} - \pi_t = (1 - \beta(1-\alpha))(p_t^* - p_t) + \beta(1-\alpha)\frac{1}{\alpha}\,\mathbb{E}_t[\pi_{t+1}].$$

The left-hand side simplifies to

$$\left(\frac{1}{\alpha} - 1\right)\pi_t = \frac{1-\alpha}{\alpha}\,\pi_t.$$

Thus

$$\frac{1-\alpha}{\alpha}\,\pi_t = (1 - \beta(1-\alpha))(p_t^* - p_t) + \beta\frac{1-\alpha}{\alpha}\,\mathbb{E}_t[\pi_{t+1}].$$

Multiply by $\frac{\alpha}{1-\alpha}$:

$$\pi_t = \frac{\alpha(1 - \beta(1-\alpha))}{1-\alpha}\,(p_t^* - p_t) + \beta\,\mathbb{E}_t[\pi_{t+1}]. \tag{11}$$

Define

$$A \equiv \frac{\alpha(1 - \beta(1-\alpha))}{1-\alpha} > 0,$$

so that

$$\pi_t = A(p_t^* - p_t) + \beta\mathbb{E}_t[\pi_{t+1}].$$

**Step 4: Substitute in the flexible-price condition.** From (7),

$$p_t^* - p_t = (\varphi - 1)y_t.$$

Substitute into (11):

$$\pi_t = A(\varphi - 1)y_t + \beta\mathbb{E}_t[\pi_{t+1}].$$

Define

$$\kappa \equiv A(\varphi - 1) = \frac{\alpha(1 - \beta(1-\alpha))}{1-\alpha}\,(\varphi - 1),$$

and we obtain the New Keynesian Phillips Curve

$$\boxed{\pi_t = \kappa y_t + \beta\mathbb{E}_t[\pi_{t+1}]}, \qquad \boxed{\kappa = \frac{\alpha(1 - \beta(1-\alpha))}{1-\alpha}\,(\varphi - 1)}.$$

## (b) Interpretation of the slope $\kappa$

The slope $\kappa$ measures how strongly current inflation reacts to the output gap.

- **Price flexibility ($\alpha$):** Higher $\alpha$ means a larger fraction of firms can reset prices in each period. For a given output gap, more firms adjust their prices, so inflation

reacts more. Hence $\kappa$ increases with $\alpha$.

- **Discount factor ($\beta$):** Higher $\beta$ means firms care more about future periods. They internalize that today's price will be in place for longer, which makes them smoother and less aggressive in their price changes for a given current output gap. This flattens the Phillips curve; $\kappa$ decreases in $\beta$.

- **Labor supply parameter ($\varphi$):** Larger $\varphi$ implies a less elastic labor supply. A given increase in output requires a larger increase in real wages and marginal costs, which raises the desired price $p_t^*$ by more. Thus the gap $p_t^* - p_t$ responds more to $y_t$, and so does inflation: $\kappa$ increases in $\varphi$.